Cell Biology

Dr. Reshna K. Albarzanji
B.Sc. M.Sc. PhD.
Cytogenetics
Hawler Medical University
College of Medicine
Microbiology dep.
reshna.kamal@hmu.edu.krd

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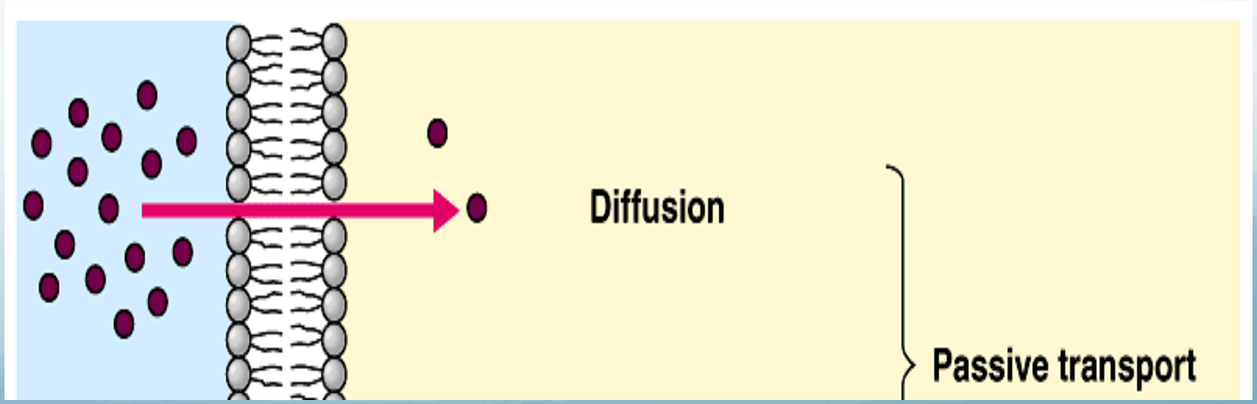
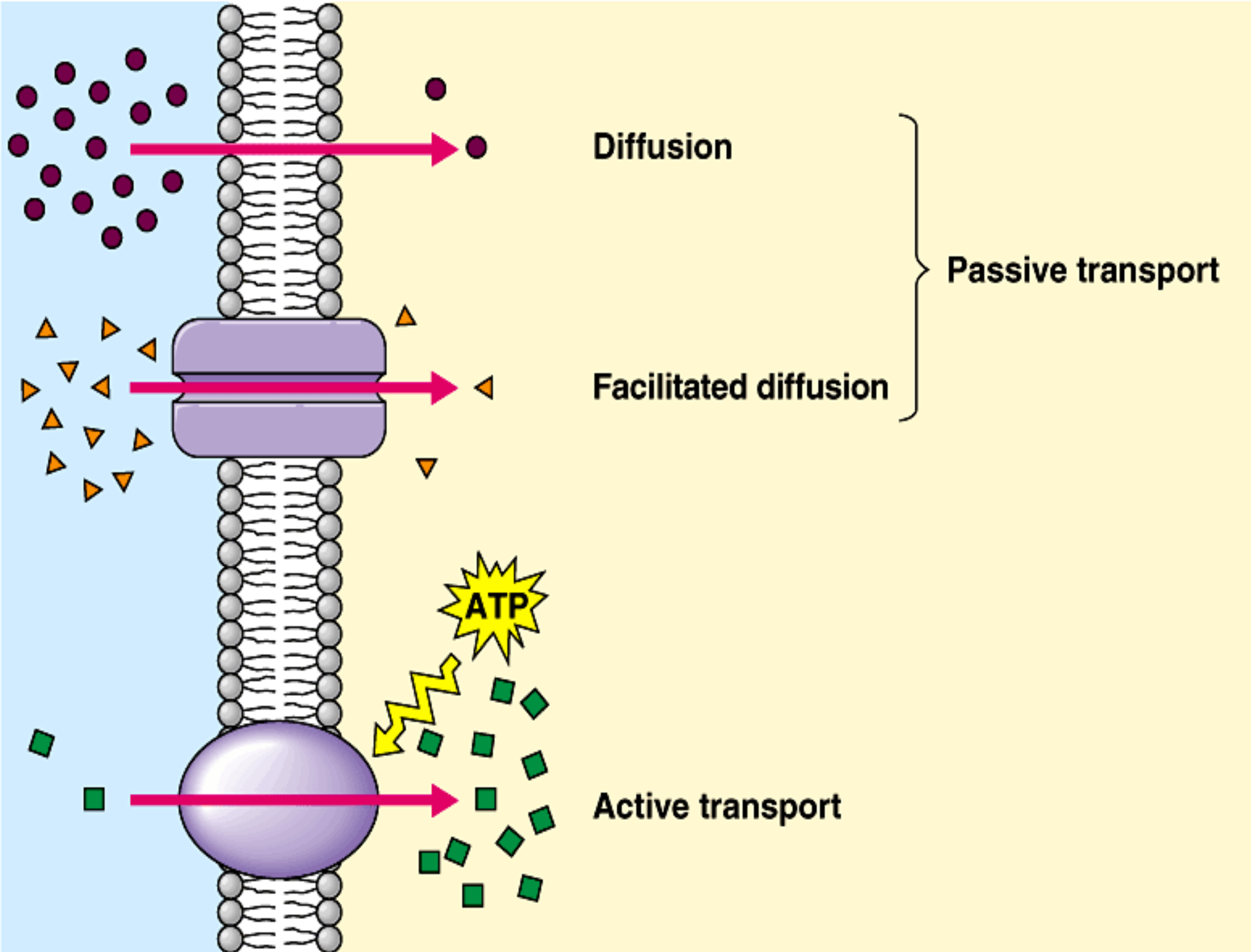
Traffic of substances through the plasma membrane

- *The traffic of substances through membranes
- 1- Passive transport:
- a- Simple diffusion.
- b-Facilitated diffusion.
- c- Osmosis.
- 2- Active transport .
- 3- Vesicular transport (Endocytosis and Exocytosis).

The traffic of substances through membranes
a-Simple diffusion:

- Diffusion is the spontaneous movement of substances down its concentration gradient and movement across membrane is called passive diffusion, during this process the molecule dissolved simply in the lipid bilayer, no protein are involved in this transport, Gases, hydrophobic molecules ex (Benzene), small polar molecules ex (Ethanol & H2O) are pass by this process.



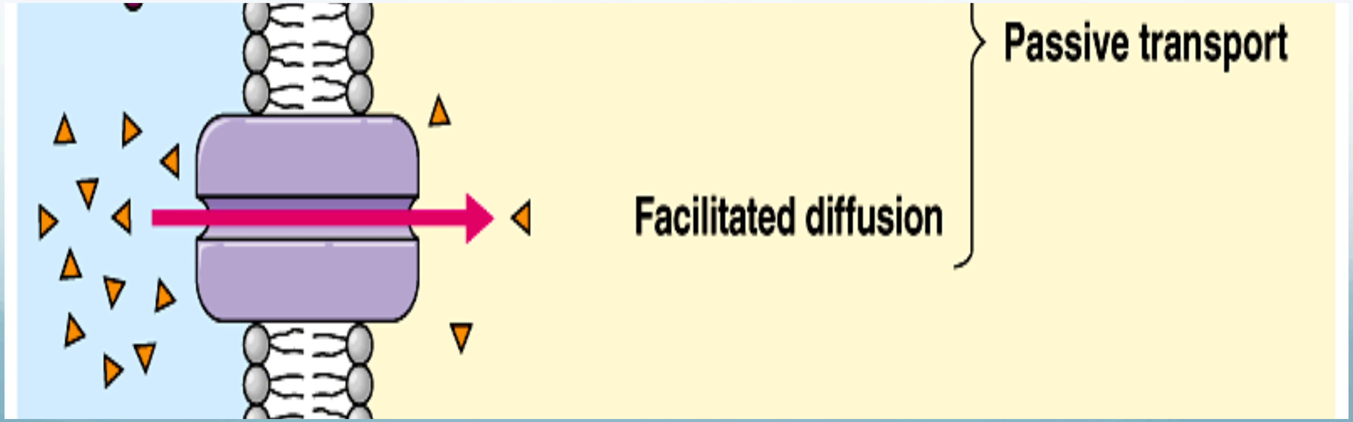

Diffusion
Passive transport
Facilitated diffusion
ATP
Active transport

b-Facilitated diffusion

- Certain molecules and ions cross the membrane with the help of transport proteins in the membrane, this movement requires no energy and its follow the concentration gradient, facilitated diffusion differs from passive diffusion in that the transport molecules do not dissolve in the phospholipid bilayer.

b-Facilitated diffusion

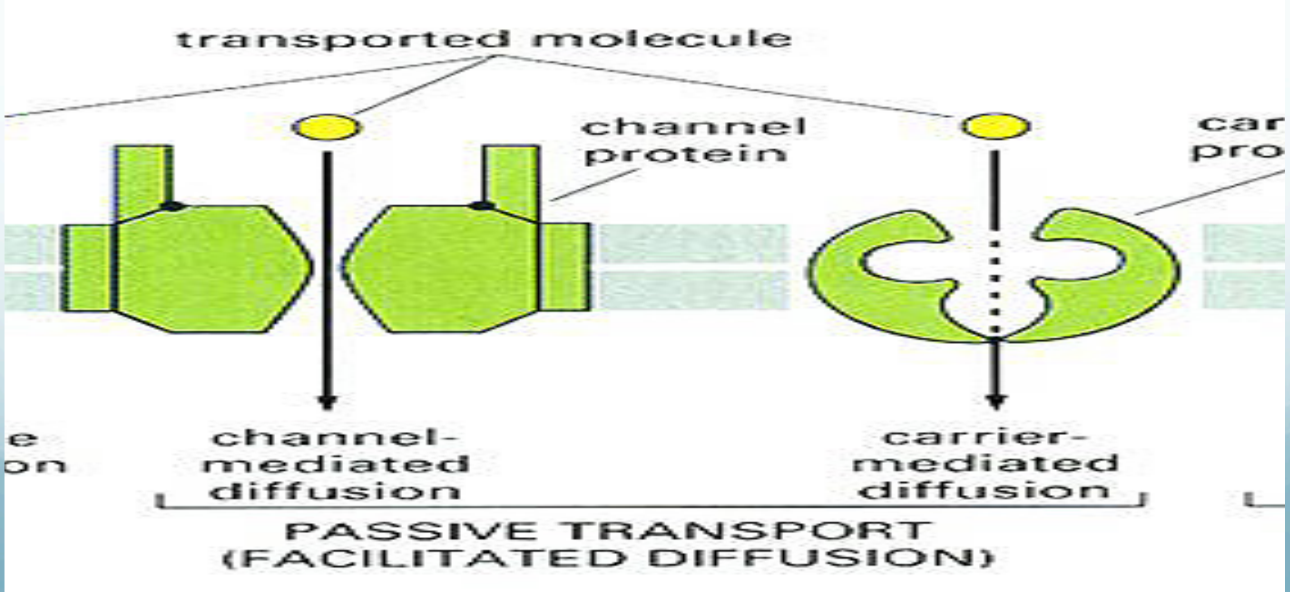
- Facilitated diffusion therefore allows polar and charged molecules such as carbohydrates, amino acids, nucleosides & ions to cross the plasma membrane.
- Two classes of proteins that mediate facilitated diffusion are generally distinguished:



b-Facilitated diffusion

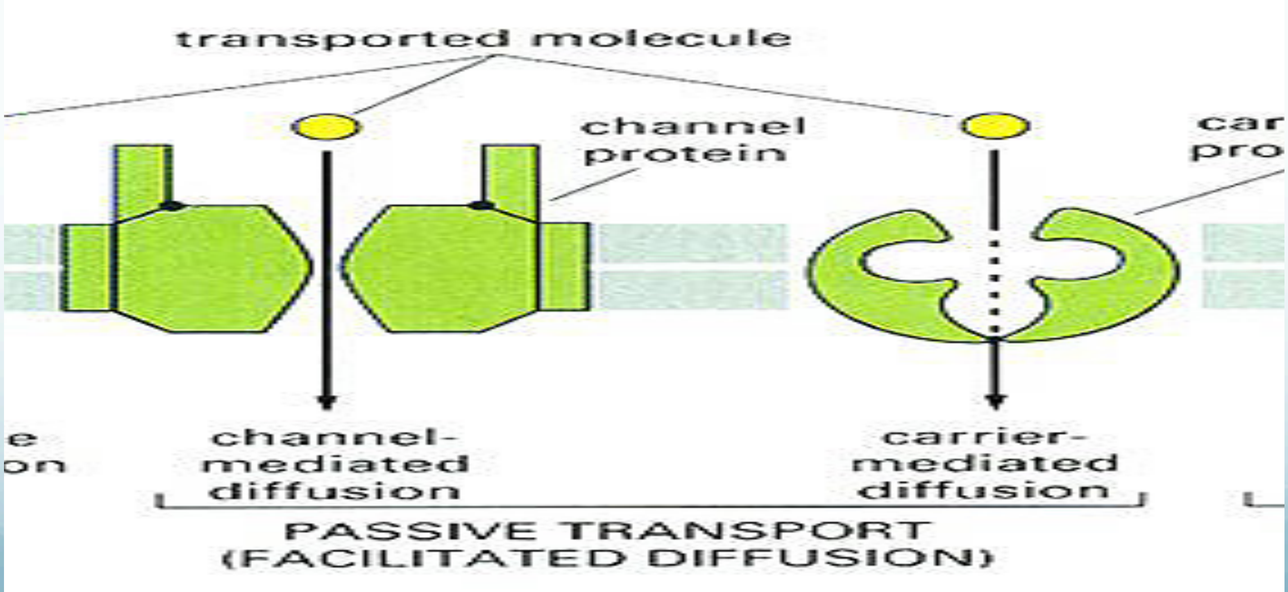
- Carrier proteins & Channel proteins.

*Carrier proteins: are responsible for facilitated diffusion of sugars, amino acids & nucleosides across the plasma membrane.



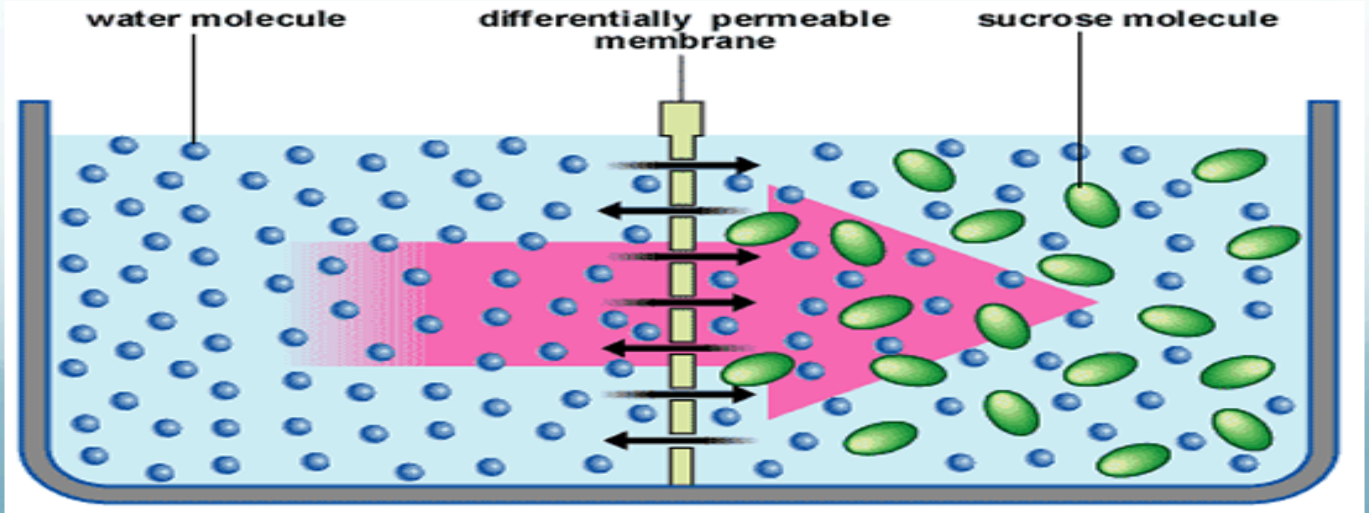
b-Facilitated diffusion

- *Channel proteins: form open pores through the membrane allowing the free diffusion of any molecule of the appropriate size and charge.



c- Osmosis

- The diffusion of water across a selectively permeable membrane.
- Water flows across a membrane from the side with a lesser concentration of solute (hypo osmotic) to the side with greater solute concentration (hyper osmotic).

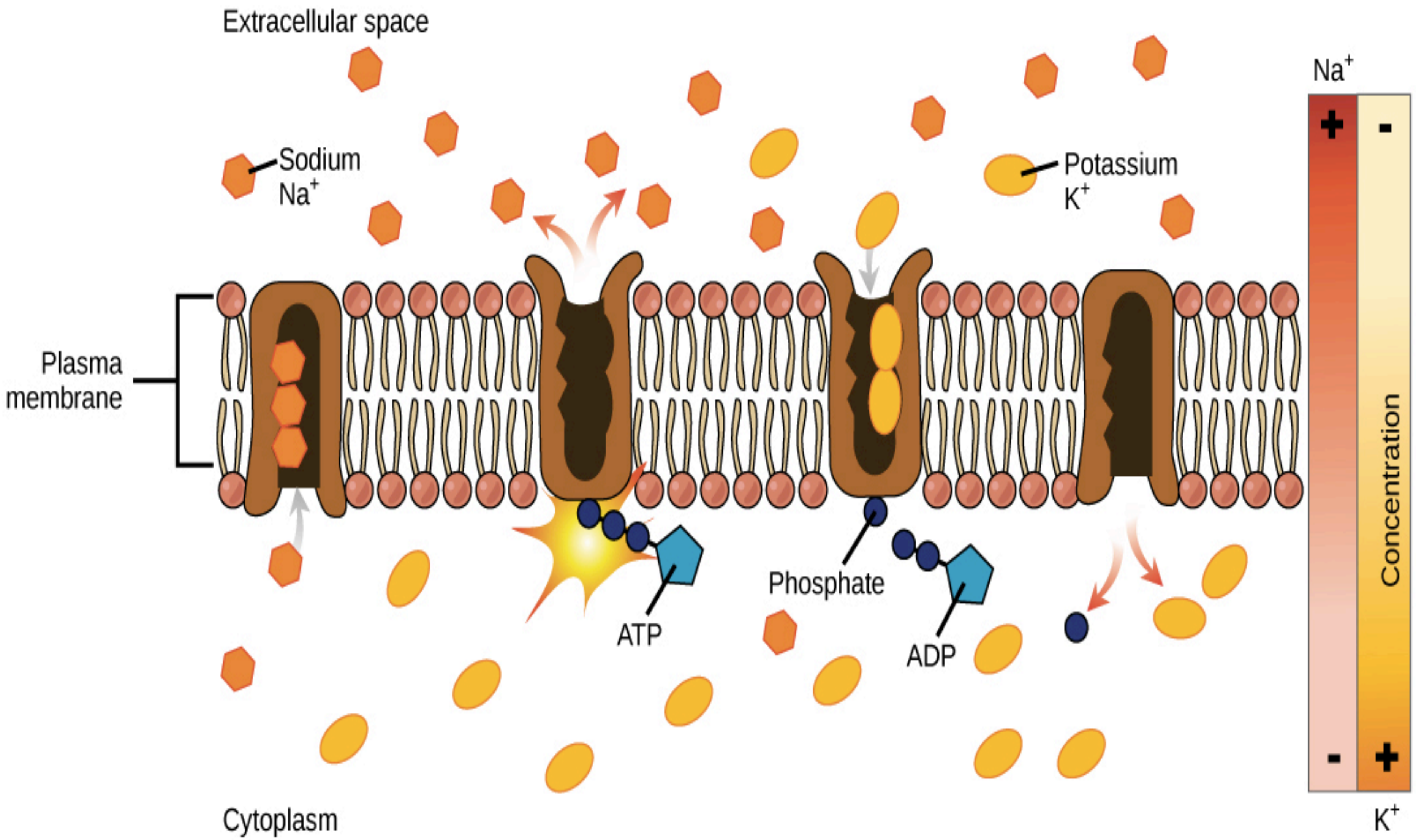


- **2-Active transport**

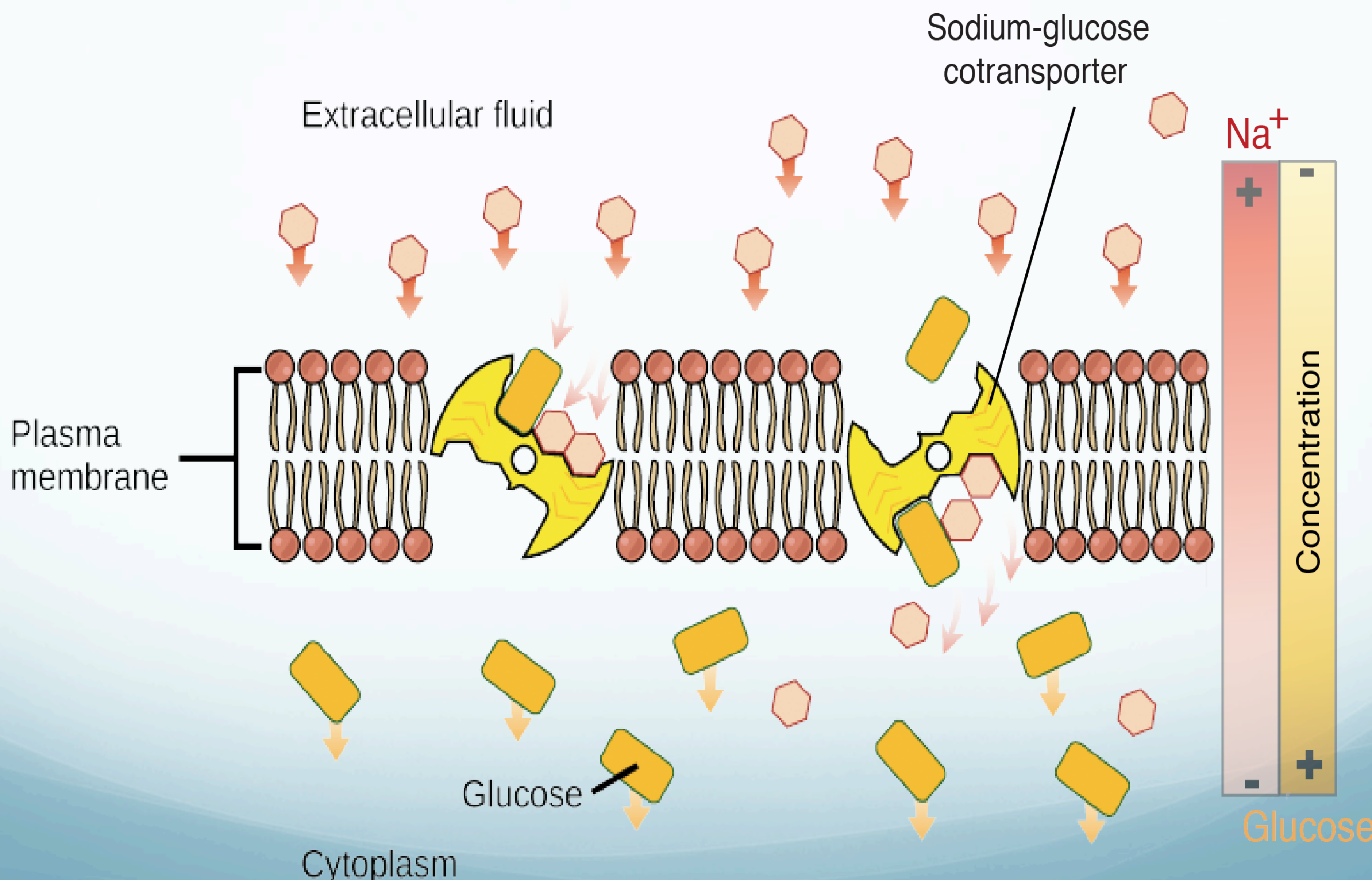
Some molecules transport against the gradient and they requires energy to pump substances across the membrane against their concentration gradient and the energy usually in ATP form.

- Active transport: is the movement of all types of molecules across a cell membrane against its concentration gradient (from low to high concentration).

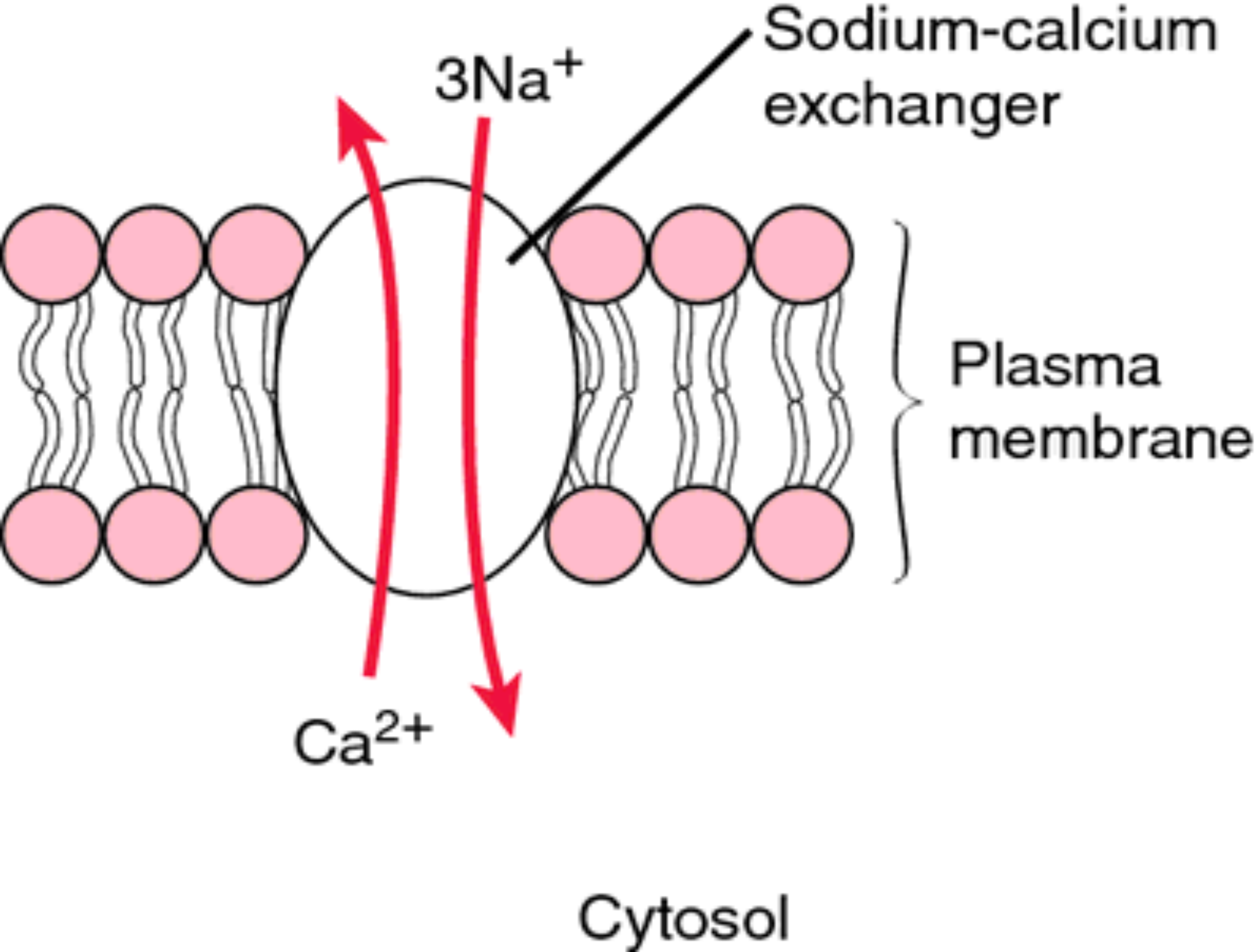
- In all cells, this is usually concerned with accumulating high concentrations of molecules that the cell needs, such as ions, glucose and amino acids. If the process uses chemical energy, such as from(ATP), it is termed **Primary active transport**
- but the type of active transport that involves the use of electrochemical gradient is called **Secondary active transport.**



[primary active transpor]



Sodium-glucose
cotransporter
Extracellular fluid
Na+
Plasma
membrane
Concentration
Glucose
Cytoplasm
Glucose



Sodium-calcium exchanger
3Na+
Plasma membrane
Ca2+
Cytosol

- **1-Primary active transport**: also called **direct active transport**, directly uses energy to transport molecules across a membrane.
- Most of the enzymes that perform this type of transport are transmembrane ATPases. A primary ATPase universal to all life is the sodium-potassium pump, which helps to maintain the cell potential.

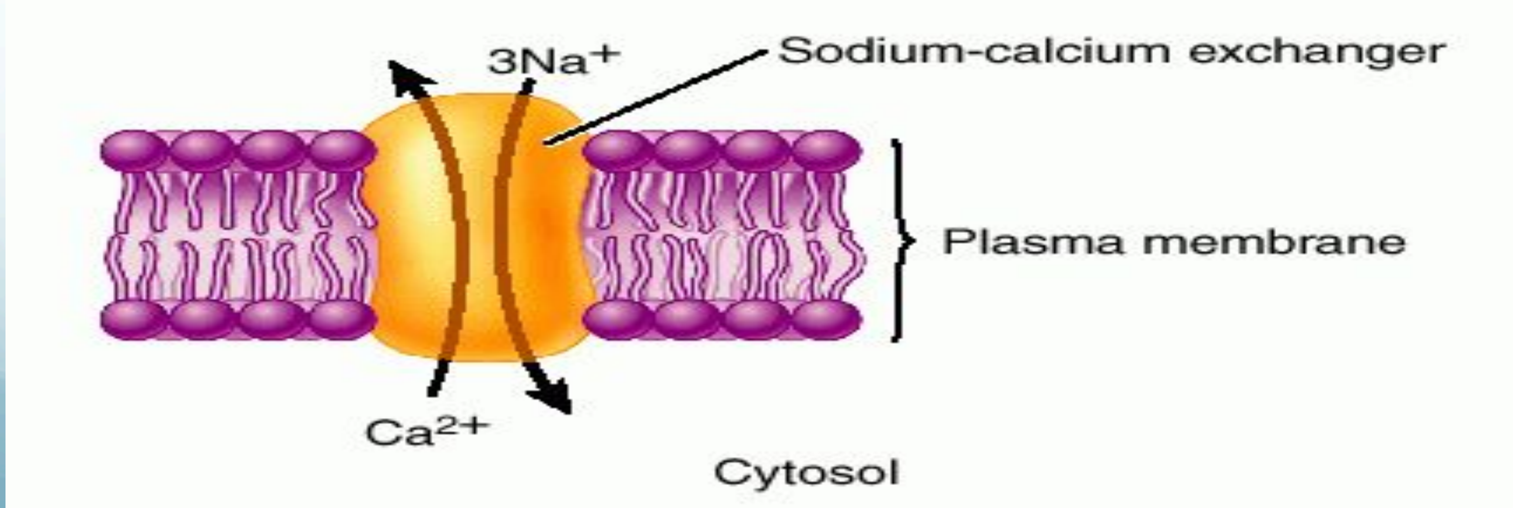
- **2-In secondary active transport**,
- also known as *coupled transport* or *co-transport*, energy is used to transport molecules across a membrane; however, in contrast to primary active transport, there is no direct coupling of ATP; instead, the electrochemical potential difference created by pumping ions out of the cell is used.
- Permitting one ion or molecule to move from the side where it is more concentrated to that where it is less concentrated increases entropy and can serve as a source of energy for metabolism. Co-transporters can be classified as symporters and antiporters depending on whether the substances move in the same or opposite directions.


Symporter
Antiporter
A
B
A
B

- **Antiport

- In an antiport two species of ion or other solutes are pumped in opposite directions across a membrane.

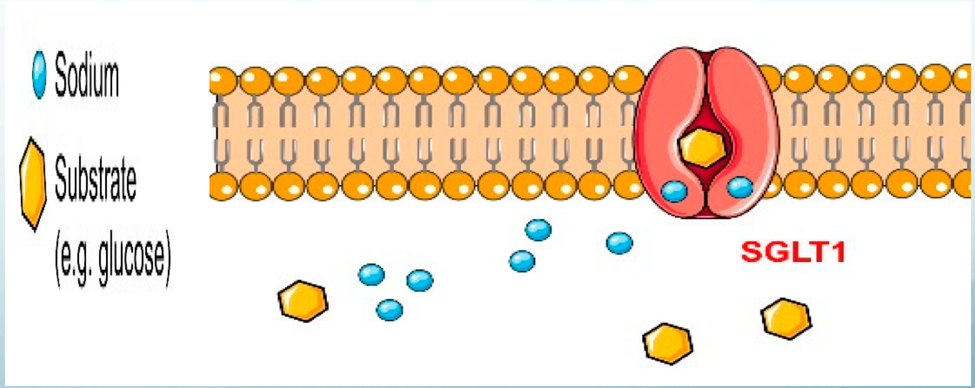
- One of these species is allowed to flow from high to low concentration which yields the entropic energy to drive the transport of the other solute from a low concentration region to a high one. An example is the **sodium-calcium exchanger** or antiporter, which allows three sodium ions into the cell to transport one calcium out



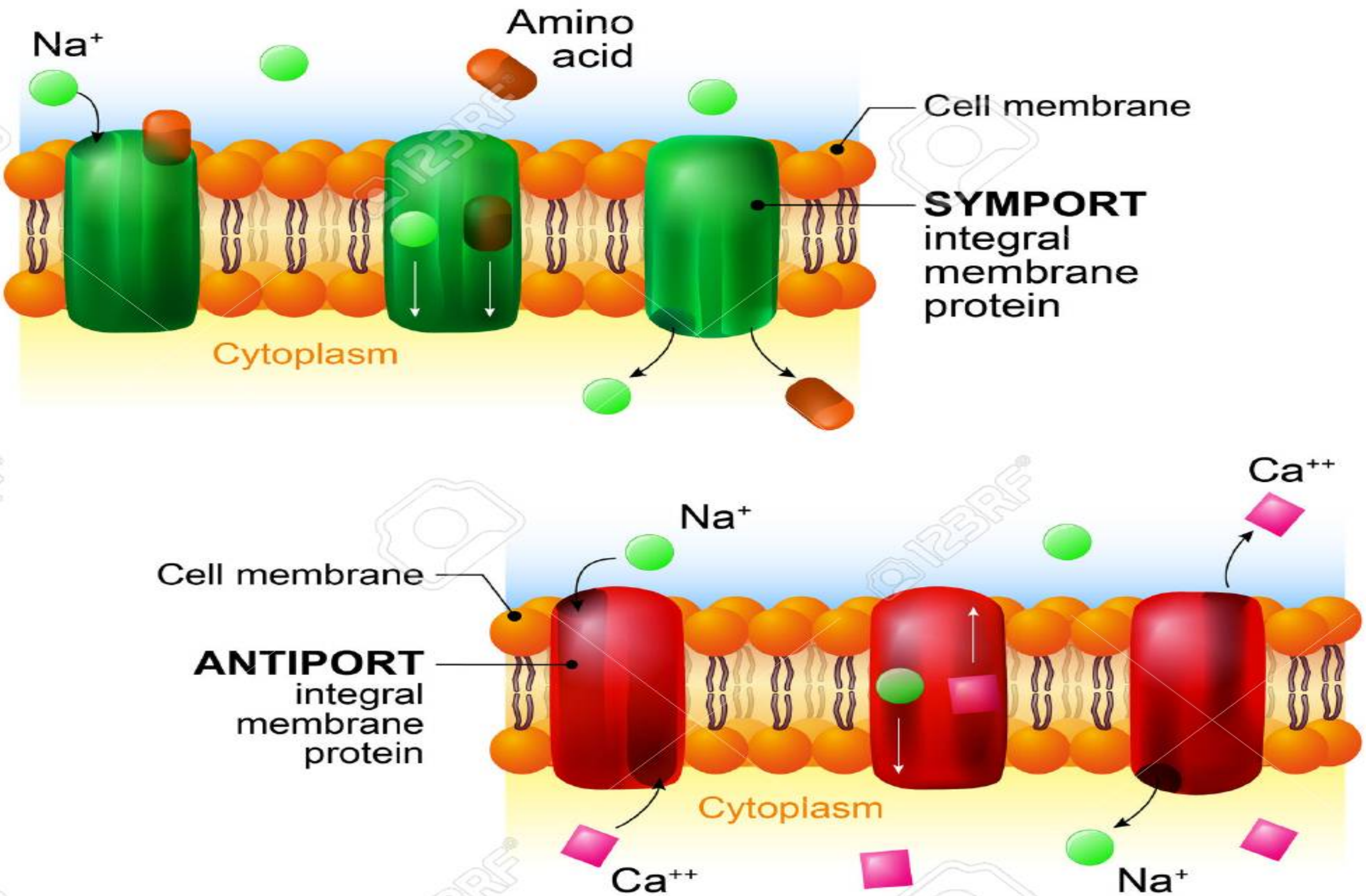
- **Symport

- Symport uses the downhill movement of one solute species from high to low concentration to move another molecule from low concentration to high concentration (against its electrochemical gradient). Both molecules are transported in the same direction.

- An example is the glucose symporter SGLT1(**S**odium/ **Gl**ucose co**T**ransporter 1), which co-transports one glucose (or galactose) molecule into the cell for every two sodium ions it imports into the cell. This symporter is located in the small intestines.

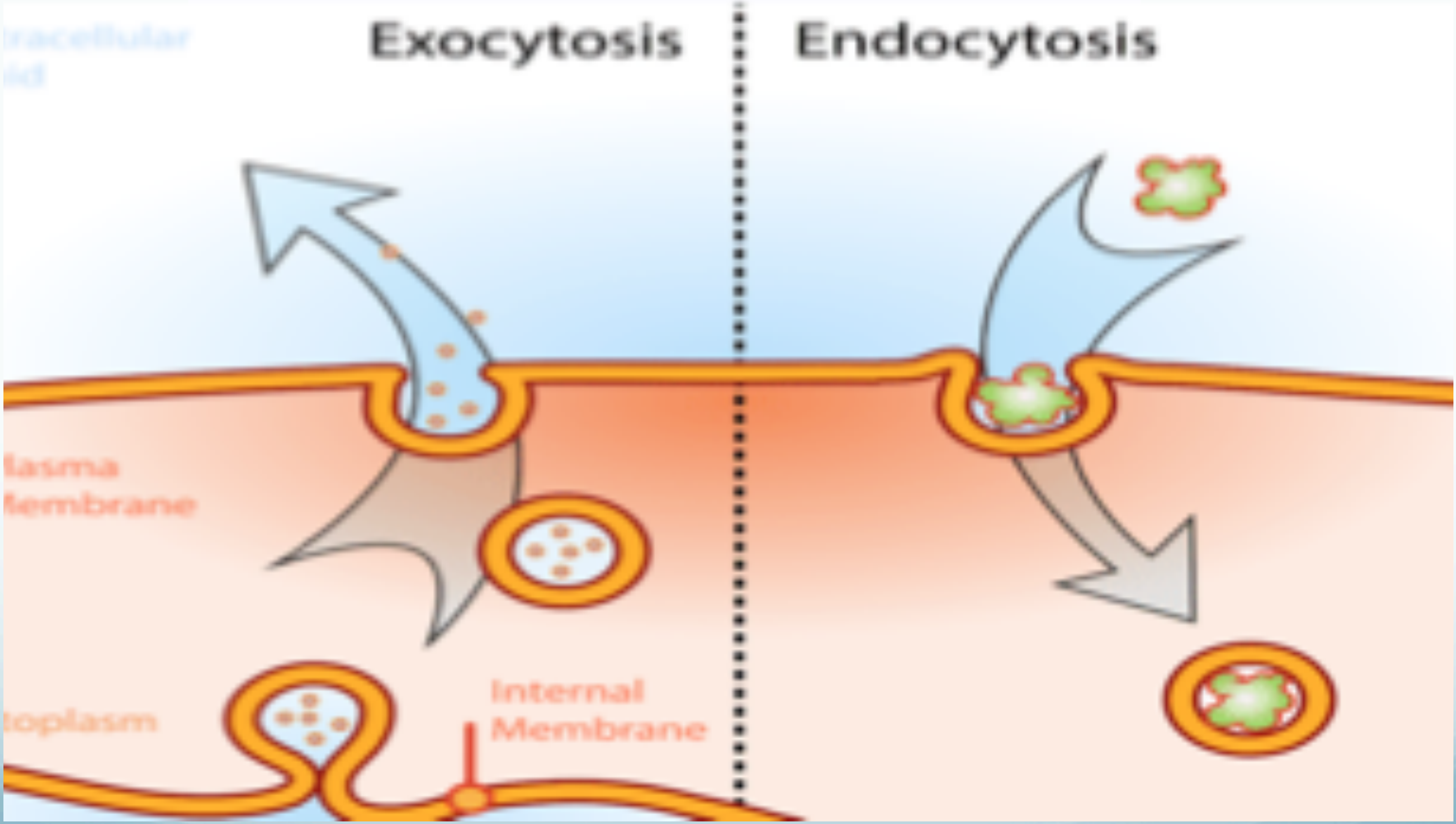


Symport & Antiport



3- Vesicle Transport

- Some molecules or particles are just too large to pass through the plasma membrane or to move through a transport protein. So cells use two other active transport processes to move these macromolecules (large molecules) into or out of the cell.
- Vesicles or other bodies in the cytoplasm move macromolecules or large particles across the plasma membrane.
- There are two types of vesicle transport, endocytosis and exocytosis Both processes are active transport processes, requiring energy.

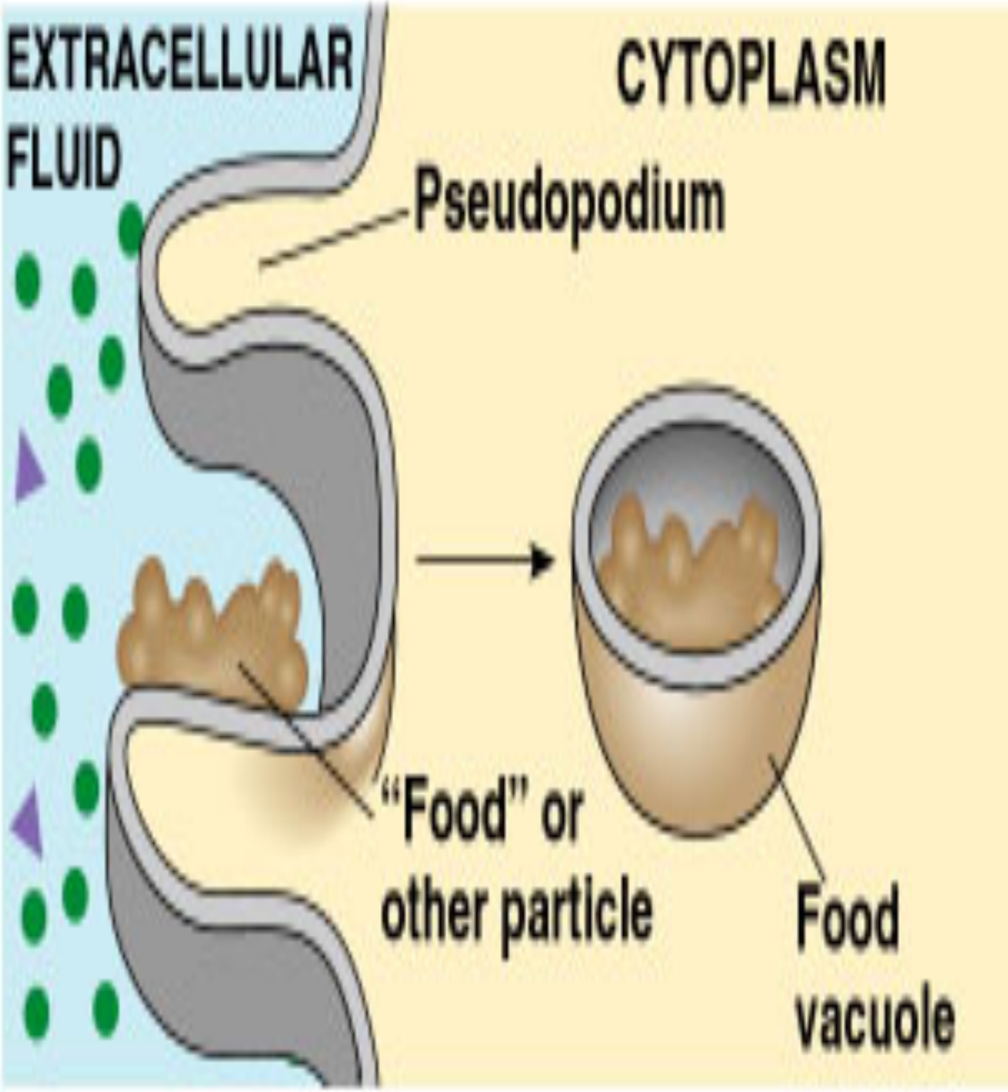

Exocytosis
Endocytosis
Internal
Membrane

- Endocytosis

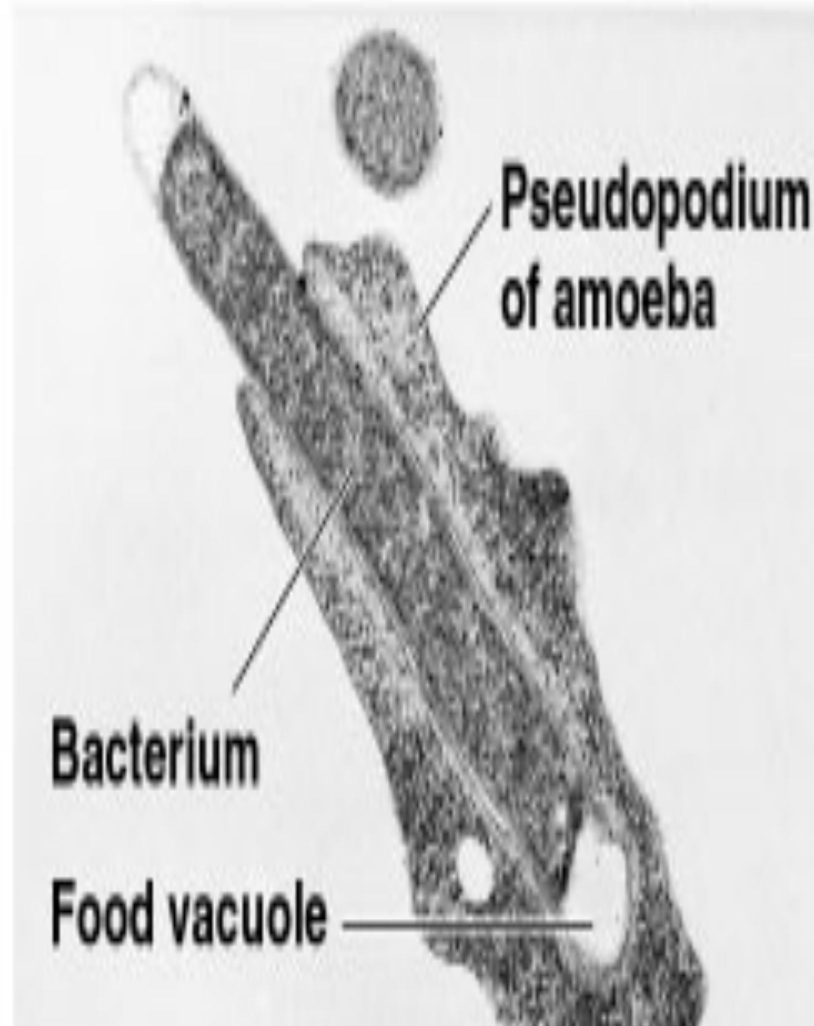
- Is the name given to the process of vesicular transport when it involves substances entering the cell.
- Endocytosis is the process of capturing a substance or particle from outside the cell by engulfing it with the cell membrane.
- The membrane folds over the substance and it becomes completely enclosed by the membrane. At this point a membrane-bound sac, or vesicle, pinches off and moves the substance into the cytosol.
- There are two main kinds of endocytosis:
- a- Phagocytosis
- b- Pinocytosis

- A-Phagocytosis : or cellular eating, occurs when the dissolved materials enter the cell. The plasma membrane engulfs the solid material, forming a phagocytic vesicle.
- Is the engulfing of large particular matter, such as micro organisms (bacteria), cell fragments such as when phagocytes engulfed defected RBCs.
-

1 µm



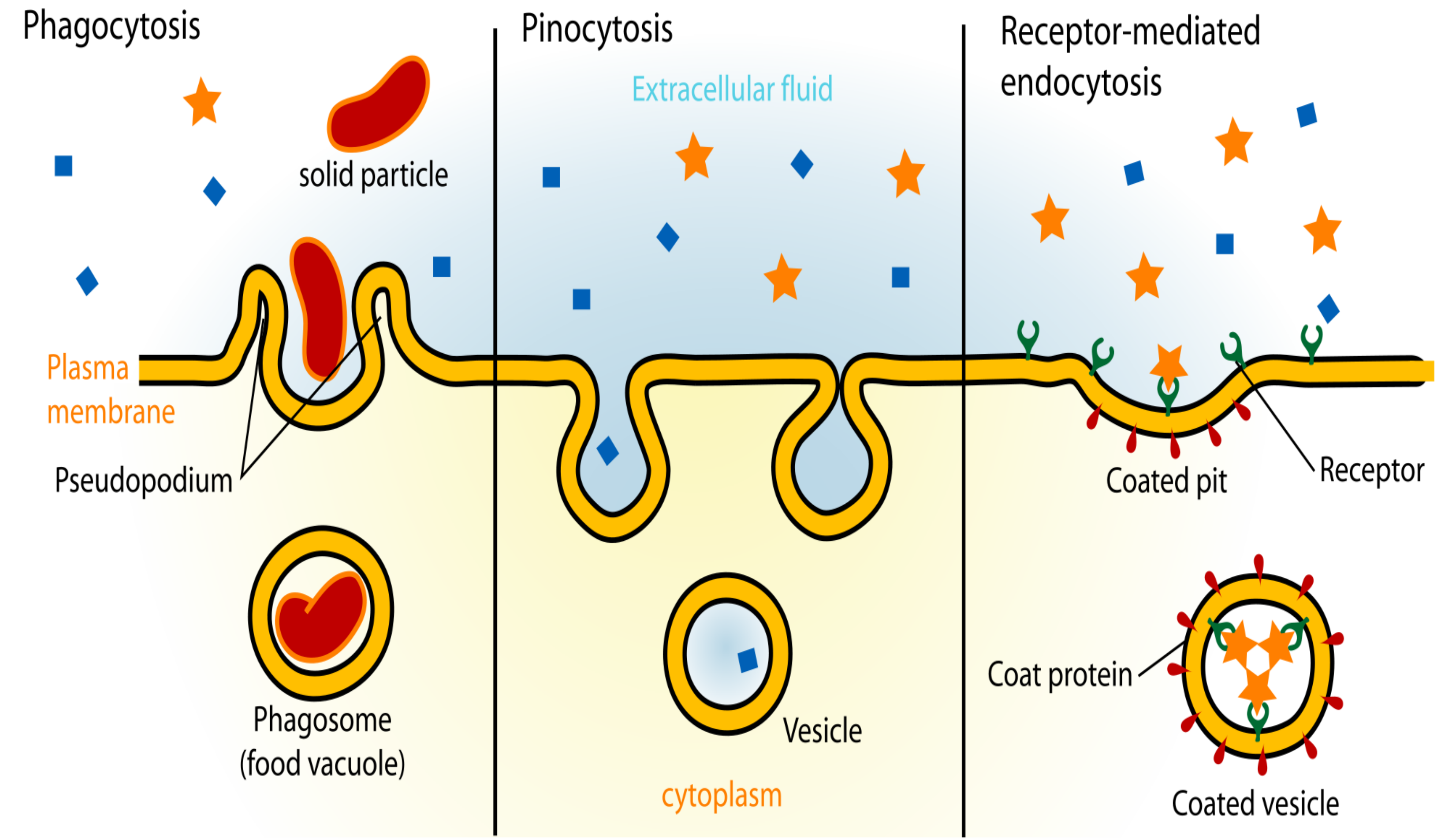
(a) Phagocytosis

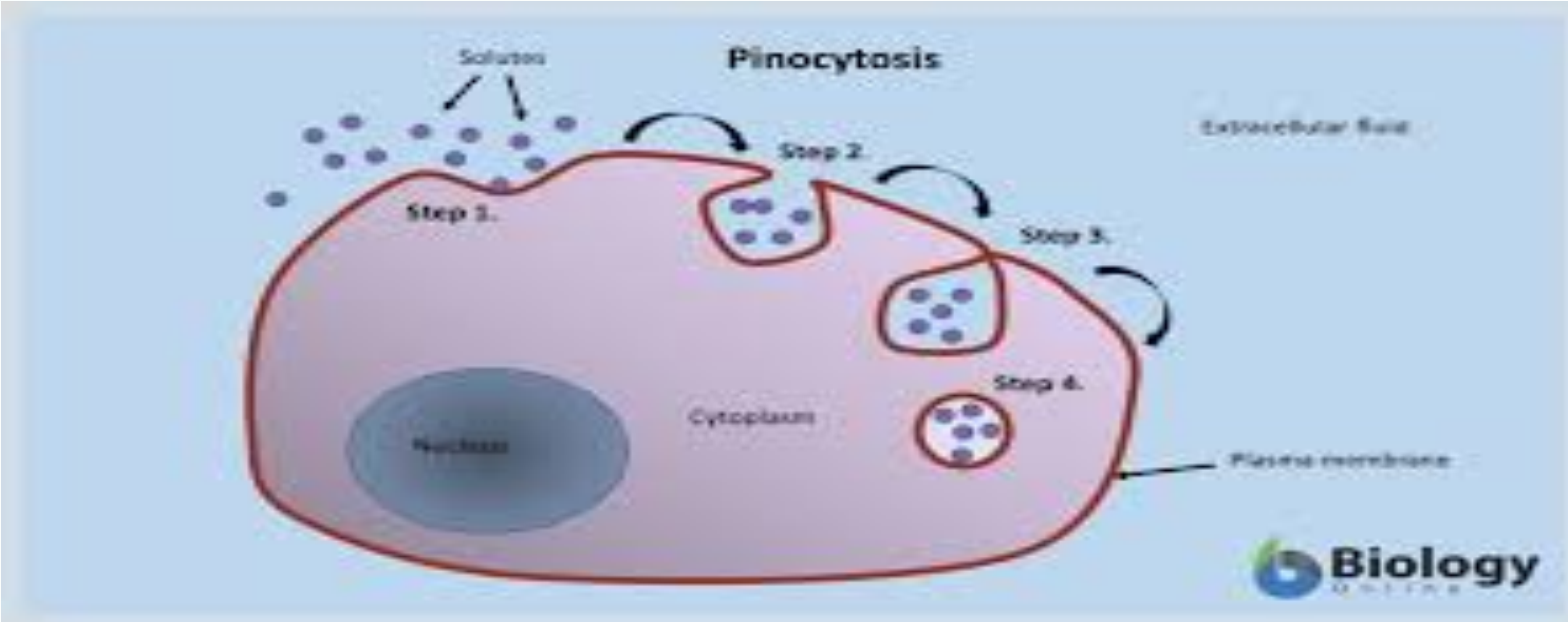


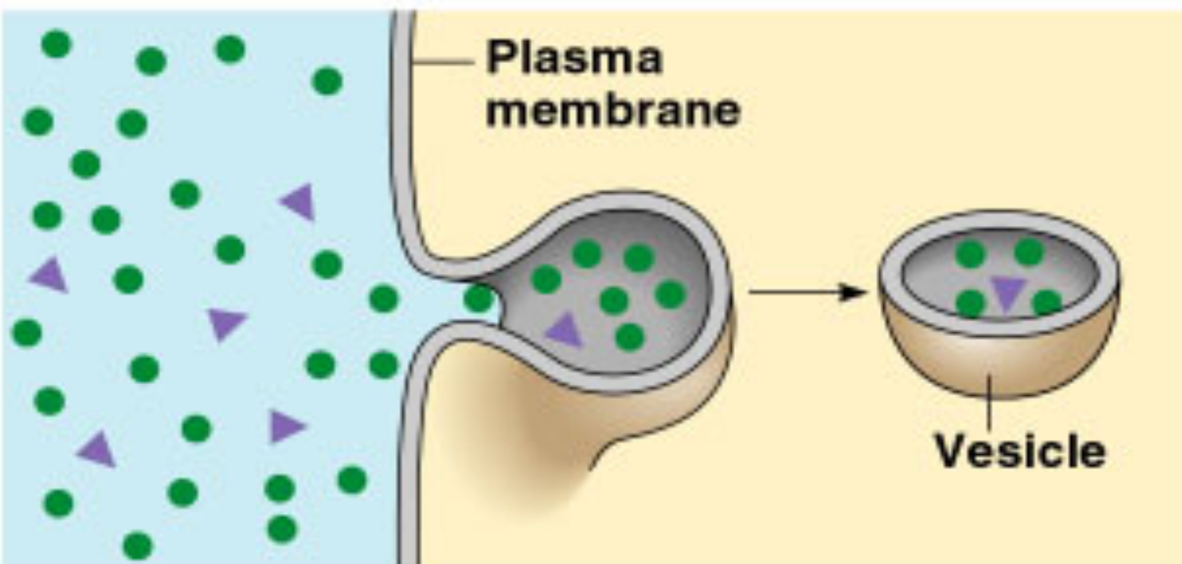
0.5 µm

- **b-Pinocytosis**, or cellular drinking, occurs when the plasma membrane folds inward to form a channel allowing dissolved substances to enter the cell. When the channel is closed, the liquid is encircled within a pinocytic vesicle.
- The term pinocytosis is formed of two words “pino” and cytosis” where ‘pino’ means “to drink” while ‘cytosis’ means relating to the cell.
- It is a continuous process in most cells and is a non-specific way for internalizing fluid and dissolved nutrients.
- The process of pinocytosis deals with the movement of a large number of tiny molecules through a fluid, which is why it is also called the fluid endocytosis or bulk-phase endocytosis.
- The molecules once inside the cells form vesicles which are then fused with the endosomes for the metabolic processes.

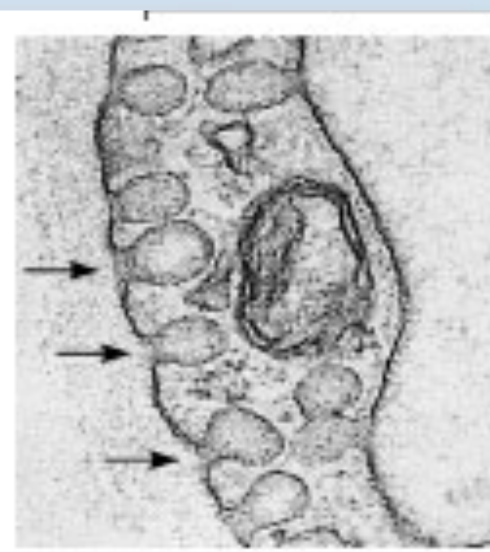
Endocytosis







(b) Pinocytosis



b-pinocytosis:

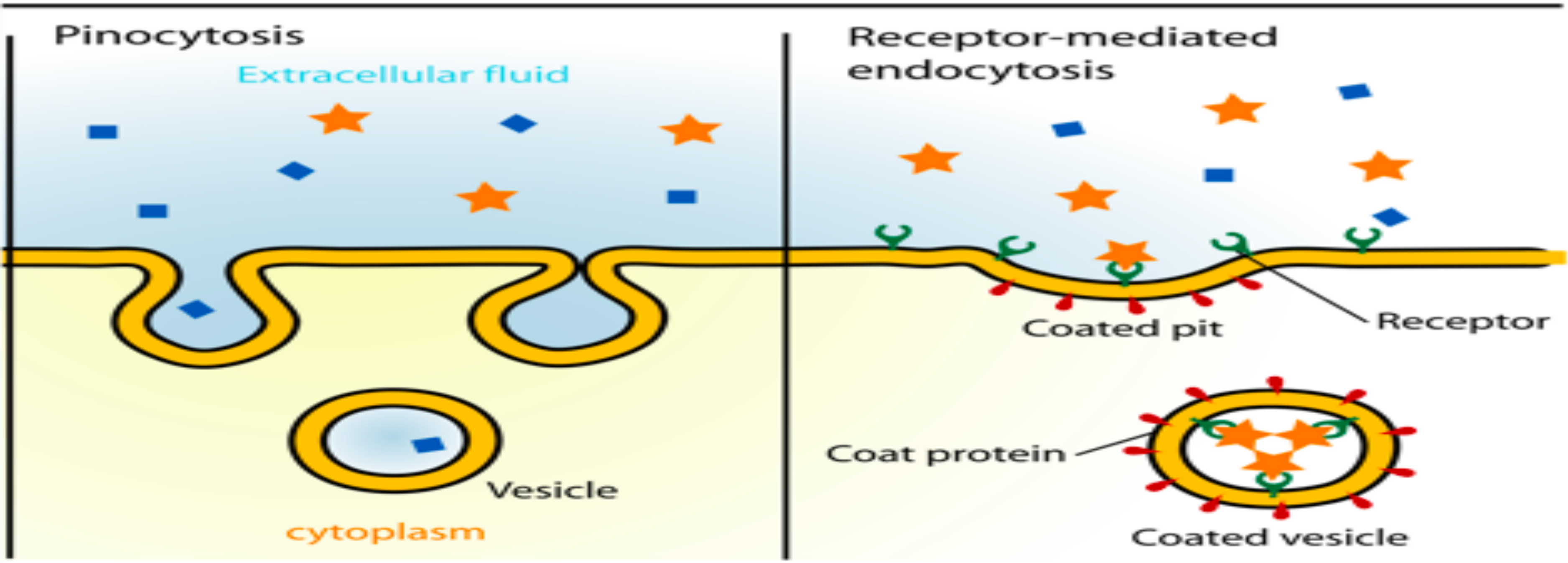
- **Pinocytosis is found in two main types**:
- **1-Smooth pinocytosis vesicles (Fluid phase endocytosis)**
- **2-Coated vesicles or (receptor mediated endocytosis)**

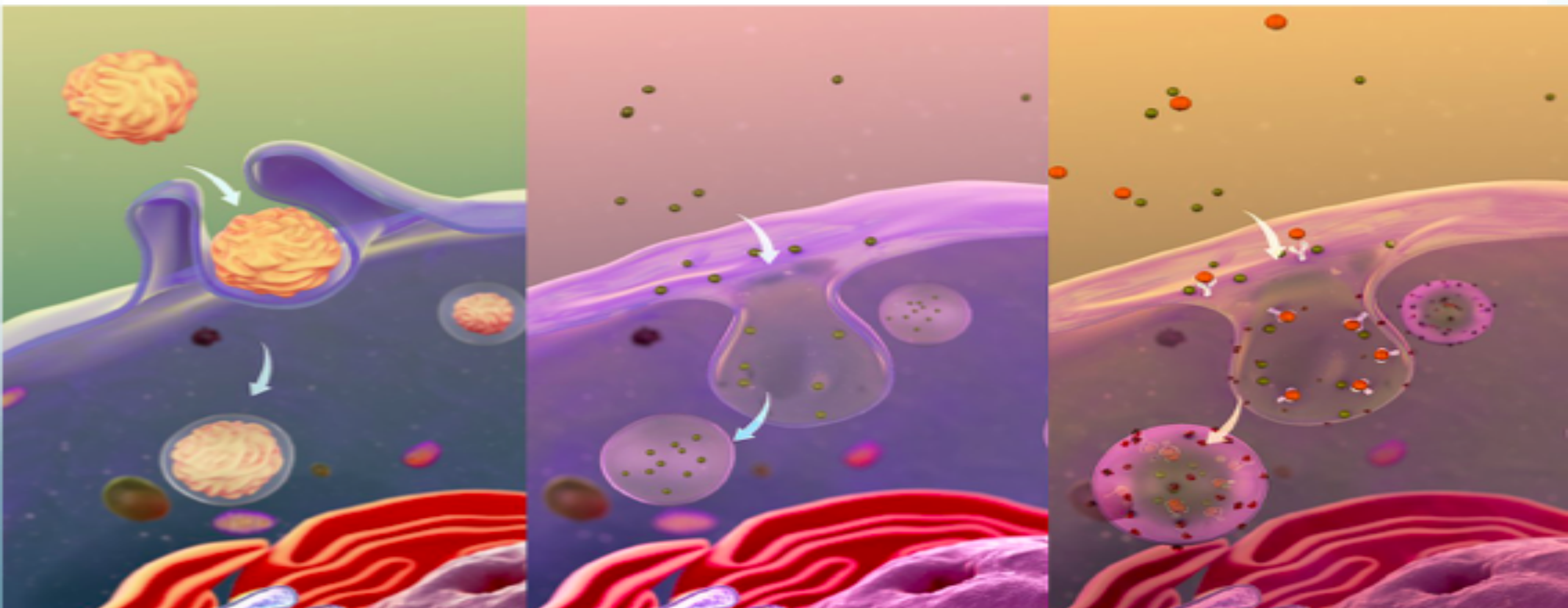
- 1-Smooth pinocytosis vesicles (Fluid phase endocytosis)
- The plasma membrane invaginate to form small pits that project into the cell the opening of the pit constricts into a narrow neck then the vesicle is separated from the plasma membrane , its numerous endothelium of blood vessels.

c-pinocytosis

C-coated vesicles or (receptor mediated endocytosis)

- The coated membrane first forms a depression then a small pit finally the coated pit pinched off, to become a coated vesicle, (receptors form any substance such as protein hormones are located at the cell surface are either dispersed or aggregated in a special region called coated pits), binding between the protein or (legend) with the receptor on the cytoplasmic membrane.

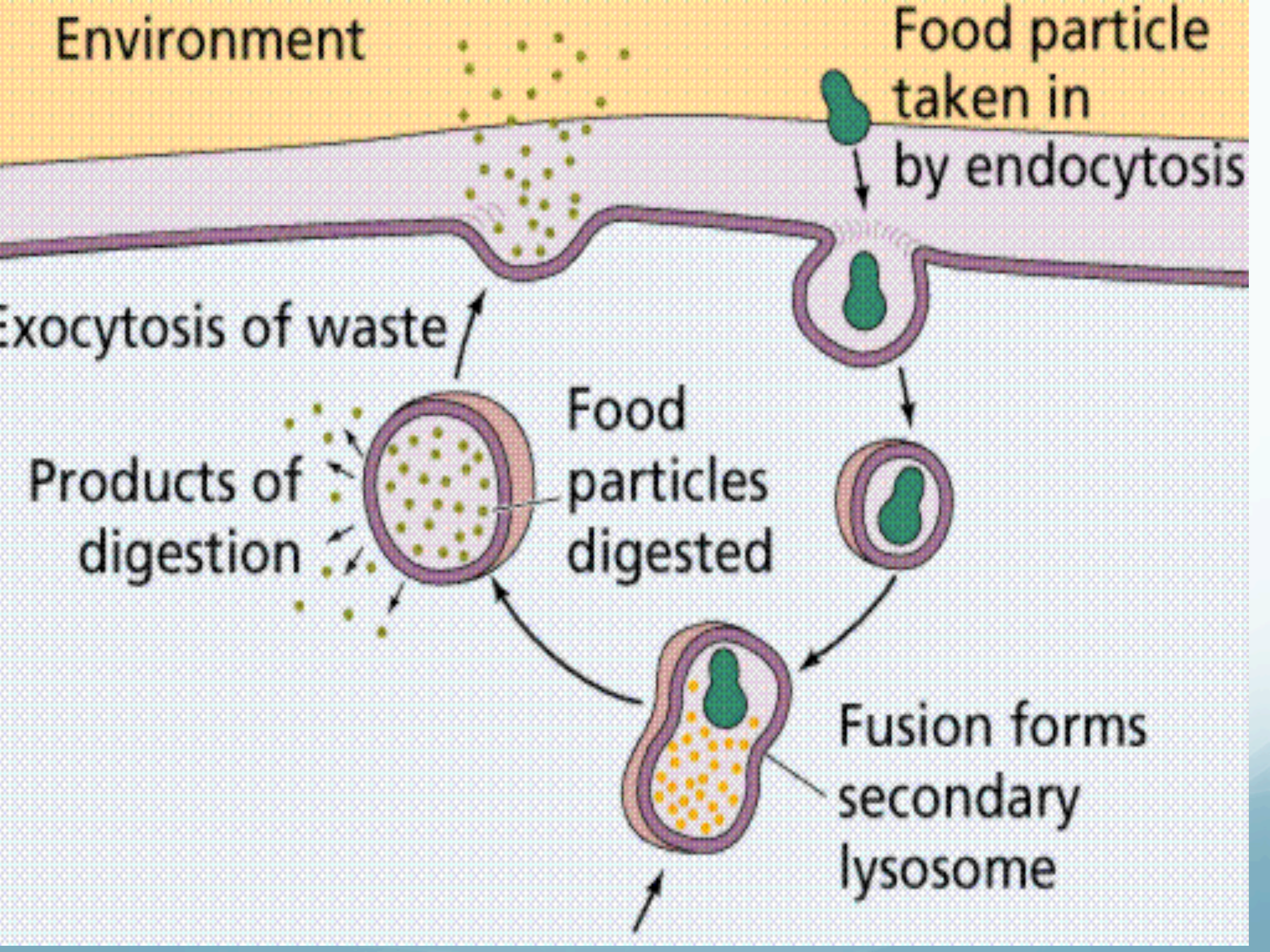

Pinocytosis
Extracellular fluid
Vesicle
cytoplasm
Receptor-mediated endocytosis
Coated pit
Receptor
Coat protein
Coated vesicle

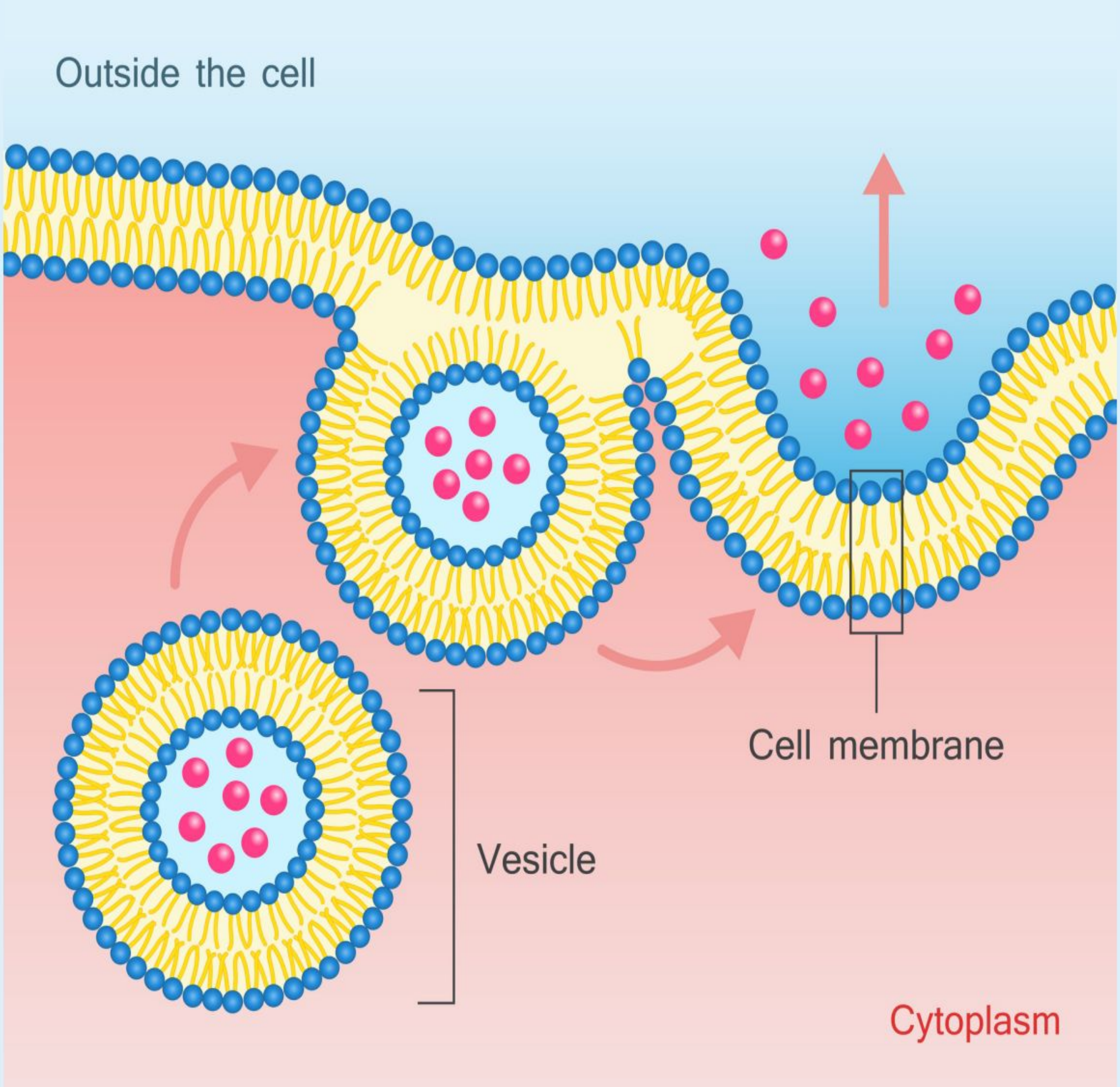


4- Exocytosis

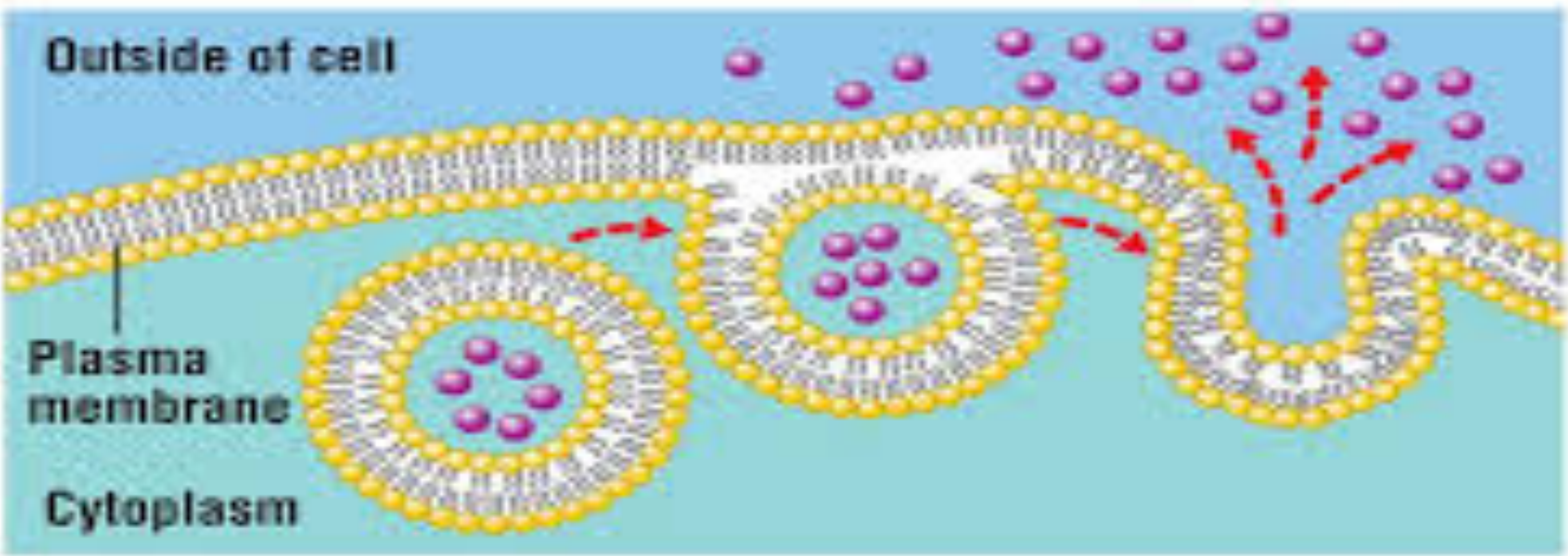
Its the name given to the vesicular trans port when it involves substances leaving the cell, large molecules are secreted by fusion of plasma membrane with the vesicles which is usually was budded from the endoplasmic reticulum or golgi apparatus .

Proteins are concentrated and stored in secretary granules then they must be activated for secretion, such as in release of zymogene granules by chief cells of gastric mucosa .


Environment
Food particle
taken in
by endocytosis
Exocytosis of waste
Food
particles
digested
Products of
digestion
Fusion forms
secondary
lysosome


Outside the cell
Cell membrane
Vesicle
Cytoplasm

- **Exocytosis**
  - Dumping molecules out of the cell (export)



(a) Exocytosis

Exocytosis and Endocytosis Introduction


Thank you

- * what are the main different types of traffic of substances through the plasma membrane?
- *What is passive transport?
- Enumerate different sub types of passive transport,

differentiate between them.

- **What is active transport?
- Enumerate different sub types of active transport,

differentiate between them.

*what is secondary active transport?

- Enumerate different sub types of secondary active transport, differentiate between them.

- *What is vesicle transport?
- Enumerate different sub types of vesicle transport, differentiate between them.
- What is coated vesicle?